Calcurion Physics Worksheets

A-Level Physics

Topic: Waves

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. State the difference between progressive and stationary waves. **[2]**

2. Define phase difference and give its unit. **[2]**

3. State the principle of superposition. **[2]**

4. Two waves of amplitude A meet in phase. What is the resultant amplitude? **[1]**

5. Define coherence. **[1]**

6. State the conditions for observable interference. **[2]**

7. In Young's double-slit, fringe spacing $w = \lambda D/s$. Define each symbol. **[3]**

8. Fringes are 1.2 mm apart when $D = 1.5$ m and $s = 0.50$ mm. Calculate λ. **[3]**

9. What is diffraction? **[1]**

10. State the single-slit diffraction minimum condition: $\sin\theta = n\lambda/a$. **[1]**

11. Light of wavelength 600 nm passes through a slit of width 0.20 mm. Calculate the angle to the first minimum. **[3]**

12. State Snell's law. **[1]**

13. Light enters glass ($n = 1.5$) from air at 40° to the normal. Calculate the refraction angle. **[3]**

14. Define critical angle and state the condition for total internal reflection. **[2]**

15. Calculate the critical angle for glass to air ($n = 1.50$). **[2]**

Section B – Hard (Q16–25)

16. Explain how a stationary wave is formed on a string. **[3]**

17. A string of length 0.80 m vibrates in its fundamental mode at 50 Hz. Calculate the wave speed. **[3]**

18. Derive the path difference for constructive interference in Young's experiment. **[3]**

19. In a diffraction grating with 500 lines per mm, first-order maximum for $\lambda = 550$ nm occurs at what angle? **[4]**

20. Explain why a blue laser produces closer fringes than a red laser in the same Young's setup. **[2]**

21. A polarising filter reduces intensity of unpolarised light to 50%. A second filter at 60° is added. Calculate the final intensity relative to the original. **[4]**

22. State Malus's law. **[1]**

23. Light of intensity I_0 passes through two polarisers with angle 30° between axes. Calculate transmitted intensity. **[3]**

24. Explain the formation of nodes and antinodes in a stationary wave. **[3]**

25. Describe an experiment to determine the wavelength of laser light using a diffraction grating. **[4]**

End of Worksheet • *Show all working clearly.*